INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

POLYSULFIDE FAIRING COMPOUND FOR WATERBORNE MAIN PROPULSION SHAFTING COVERINGS ON U.S. NAVAL SURFACE SHIPS AND SUBMARINES

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) describes the fairing/sealing compound that covers and protects all shaft-sleeve interface areas of waterborne propulsion shafting by preventing galvanic corrosion to the steel substrate in the marine environment and so preventing structural failure of the shafting. This CID pertains specifically to the fourth component of the five-component shaft covering system specified in MIL-STD-2199.

2. CLASSIFICATION. The layers shall conform to the following types, as specified (see 7.2):

a. Type I: First layer (primer). Low-viscosity primer for polysulfides.

b. Type II: Deleted.

c. Type III: Fourth component (sealing/fairing compound). High-viscosity, thixotropic elastomeric coating.

3. SALIENT CHARACTERISTICS.

3.1 First component (primer). The first layer of the shaft covering system shall consist of a low-viscosity primer for polysulfides, for use only when applied to bare substrate (e.g., Inconel®, or equal, cladding). The primer shall contain rubber and a coupling agent in solvent (toluene) that promotes adhesion of the fourth component (elastomeric fairing compound) of the waterborne propulsion shaft covering system by delivering a monolayer of rubber to the media blasted carbon steel substrate. The primer (Thiokol 415 or low-volatile organic compound [VOC], two-component Thiokol 5050, or equal) shall be specifically intended for use as a primer over steel for polysulfide coatings. The primer is the only shaft covering component that may contain (VOCs) (see 4.2).

3.1.1 Adhesion of primer. Adhesion of the primer to steel (or Inconel®, or equal), and adhesion of the primer to the elastomeric fairing compound, shall be demonstrated both dry and after 1-year immersion in seawater in accordance with ASTM D1141. The primer shall be applied to flat steel test panels, 12 by 12 by ¼ inches in accordance with MIL-R-23461, that have been white metal blasted to a 3 mil average profile in accordance with SSPC-SP 5/NACE No. 1. After the primer dries, the elastomeric coating specified in 3.2 shall be applied to a ⅛-inch thickness in accordance with MIL-STD-2199. After the elastomeric coating cures, it shall be covered with two plies of glass reinforced plastic (GRP) in accordance with MIL-STD-2199. Primer adhesion shall be determined in accordance with ASTM D4541 using a portable adhesion tester such as the P.A.T.T.I.® (Pneumatic Adhesion Tensile Testing Instrument), or equal. The P.A.T.T.I.® tester, or equal, shall be applied to the GRP. Both dry and wet primer adhesive strength values shall exceed 400 pounds per square inch (psi). Adhesive failure at the primer-to-steel bond or elastomeric coating to primer bond shall constitute rejection of the primer.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 9330

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3.1.2 Drying characteristics. The primer shall dry in 16 hours or less.

3.1.3 Application characteristics. The primer shall be capable of application by roller and shall have a minimum coverage of 400 square feet per gallon. Coated areas shall be identifiable from uncoated areas by a transition in gloss.

3.1.4 Corrosion protection performance. The primer shall contain a polymer, in addition to adhesion promoters, to protect the steel substrate from corrosion. The ability of the primer to protect the steel from corrosion shall be demonstrated by applying the primer within 4 hours to half of a media blasted 4- by 4-inch carbon steel alloy test coupon in accordance with ASTM A36/A36M. The remaining half of the test coupon shall be left uncoated. The coupon shall then be immersed in artificial seawater prepared in accordance with ASTM D1141. The test coupon shall be removed from the water weekly, photographed, and reimmersed in seawater in accordance with ASTM D1141. The coated-half of the test coupon shall remain free of corrosion after 1-month immersion in seawater in accordance with ASTM D1141.

3.2 Fourth component (sealing/fairing compound). The fourth component of the shaft covering system shall be an elastomeric sealing/fairing compound, specifically, a polysulfide with a mixed viscosity of 700,000 centipoise or greater (Thiokol 2282, or equal). The sealing/fairing compound shall consist of a 100 percent solid, high viscosity, thixotropic elastomeric coating for fairing the shaft to the sleeve and as a filler material for surface irregularities in the shaft. The sealing/fairing compound shall bond to the ultra-high solids (UHS) coating and to the GRP shell specified in MIL-STD-2199.

3.2.1 Adhesion of sealing/fairing compound to MIL-PRF-23236, type VII UHS epoxy amine coatings; 4 to 8 mils of Sherwin-Williams Fast Clad Epoxy Primer (or equal) followed by 18 to 22 mils of Sherwin-Williams Fast Clad ER Epoxy (or equal). Adhesion of the sealing/fairing compound to the UHS coating shall be tested both dry and after 1 year immersion in seawater in accordance with ASTM D1141. Adhesion shall be determined in accordance with ASTM D4541 with the P.A.T.T.I.®, or equal, portable tester applied to the GRP or in accordance with MIL-R-23461. The second and third components of MIL-STD-2199 shall be applied to 12- by 12- by ¼-inch flat steel test panels in accordance with MIL-R-23461 that have been white metal blasted to a 3-mil average profile in accordance with SSPC-SP 5/NACE No. 1. Primed and coated test panels shall be covered with a 20±5-mil (0.020±0.005-inch) layer of sealing/fairing compound after the UHS coating cures. After the sealing/fairing compound cures, the test panel shall be covered with four plies of GRP applied in accordance with MIL-STD-2199. Both dry and wet sealing/fairing compound to UHS coating adhesive strength values shall exceed 400 psi and any failure shall occur cohesively in the sealing/fairing compound. Adhesive failure at the sealing/fairing compound-to-UHS coating bond shall constitute rejection of the sealing/fairing compound.

3.2.2 Adhesion of GRP to sealing/fairing compound. Adhesion of GRP to the sealing/fairing compound shall be tested both dry and after 1-year immersion in seawater in accordance with ASTM D1141. Primed and coated test panels specified in 3.2.1 shall be covered with a 20±5-mil (0.020±0.005-inch) layer of sealing/fairing compound after the coating cures. After the sealing/fairing compound cures, the test panel shall be covered with four plies of GRP applied in accordance with MIL-STD-2199. GRP adhesion to the sealing/fairing compound shall be determined in accordance with ASTM D4541 using a portable adhesion tester such as the P.A.T.T.I.®, or equal. The P.A.T.T.I.® tester, or equal, shall be applied to the GRP. Both dry and wet GRP-to-sealing/fairing compound adhesive strength values shall exceed 400 psi, and any failure shall occur cohesively in the sealing/fairing compound. Adhesive failure at the GRP-to-sealing/fairing compound bond shall constitute rejection of the sealing/fairing compound.

3.2.3 Cure characteristics. The sealing/fairing compound shall achieve full cure at ambient room temperature. The pot life shall be a minimum of 30 minutes.

3.2.4 Curing constituents. The polysulfide components of the sealing/fairing compound shall be free of solvents and lead oxide curing agents.

3.2.5 Application characteristics. The sealing/fairing compound shall be capable of being applied to rotating shaft sections using a trowel or squeegee. The viscosity or thixotropy shall allow the material to fair the shaft to the sleeve end.

3.2.6 Solvent content. The sealing/fairing compound system shall be free of solvents and volatile organic compounds.

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3.2.7 Tensile properties. Tensile elongation of the cured sealing/fairing compound shall exceed 400 percent at failure when tested in accordance with ATSM D638.

3.3 Toxicity. The sealing/fairing compound shall have no adverse effect on the health of personnel when used for its intended purpose and shall not be a hazard to the environment during waste disposal.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental Protection Agency (EPA) compliancy. The amount of VOCs released shall be in accordance with the limits established by the EPA for shipyards in 40 CFR 63.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered shall have been previously sold either to the government or on the commercial market and shall currently remain a standard (currently produced) offering of the manufacturer. A currently produced item is required to help ensure up-to-date technology, especially in light of probable multiple-year contract durations and rapid advancements experienced in the technology.

6. PACKAGING.

6.1 First layer preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as "Propeller Shaft Fairing Compound Primer".

6.2 Fourth component preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as "Propeller Shaft Sealing/Fairing Compound".

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.1.2 Code of Federal Regulations (CFR). The CFR may be obtained from the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401 or online at www.ecfr.gov.

7.1.3 Defense specifications and standards. Defense specifications and standards are available online at https://quicksearch.dla.mil.

7.1.4 Federal Acquisition Regulation (FAR). The FAR may be obtained from the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

7.1.5 Society for Protective Coatings (SSPC) standards. SSPC standards may be obtained from SSPC Publication Sales, 40 24th Street, 6th Floor, Pittsburgh, PA 15222-4656 or online at www.sspc.org.

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7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Type (see 2).
c. Product conformance provisions (see 5.1).
d. Packaging requirements (see 6).
e. When Safety Data Sheets (SDS) are required (see 7.4).
f. Quantity required.

7.3 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances in accordance with the conversion tables specified in FED-STD-376 and all other requirements of this CID are met, including form, fit, and function. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.4 SDS. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313. 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the SDS.

7.5 Key words.

Elastomer
Flexible sealants
Polysulfide coatings
Polysulfide sealant

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – MR
Navy – SH
Air Force – 99

Review Activities:
Navy – AS, CG
Air Force – 03, 11, 84

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 9330-2019-002)

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